VINCENT VU

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EDUCATION

San Jose State University [ABET Accredited] — **San Jose, California**
M.S Mechanical Engineering — *Expected: December 2028*
B.S Mechanical Engineering — *May 2026*

RELEVANT SKILLS

Design/Testing: SOLIDWORKS, Siemens NX, Design for Manufacturing & Assembly, Design of Experiments, ANSI/ASME Drawing Standards, GD&T, Fits/Tolerances, Finite Element Analysis, Electronics Assembly and Testing, Plastic Part Design, Metal Part Design
Fabrication: 3D Printing, CNC Router, Laser Cutter, Mill, Lathe, Power and Hand Tools, Welding, Wiring/Soldering, Woodworking
Software: MATLAB, Arduino, Raspberry Pi, ANSYS, LabVIEW, LTspice, Visual Studio Code, Adobe Creative Suite, Microsoft Office

PROFESSIONAL EXPERIENCE

Tesla – Hardware Validation Engineering Technician — **July 2026 – October 2026**
Optimus Test Team (OTT) — *Palo Alto, California*

- Built a standardized defect-reporting workflow (hardware failure modes), cutting inspection and reporting time by 50%
- Identified 10+ hardware failure modes through component teardown, inspection, and fixture-based testing
- Improved engineering handoffs by preventing 2–5 duplicate failure tickets per test through centralized issue tracking

Amazon – Area Manager Intern — **June 2025 – August 2025**
Fulfillment Operations | Multi-Item Orders — *Eastvale, California*

- Reduced multi-item order defects by 40% through ATLAS root-cause analysis and targeted process improvements
- Increased multi-item order fulfillment by ~10% by implementing visual indicators to prevent oversized-item holds

PROJECTS

Office Stapler Remover Redesign - Sheet-Metal Press — **March 2026**

- Reduced staple-removal actions by 67% through user research and iterative mechanism design in SOLIDWORKS
- Predicted 18% lower actuation force through a 4-factor Design of Experiments
- Analyzed 4 critical interfaces using tolerance analysis to target 0.05–0.55 mm functional clearances
- Designed an 8-station progressive-die concept and 3D-printed a tooling demonstrator to learn die fundamentals
- Simulate 25,000 loading cycles by comparing ANSYS predictions with physical mechanism testing

FinderBot - Plastic Injection Molding — **May 2026**

- Developed an autonomous item-finding rover integrating AI vision, servo actuation, and electronic controls
- Packaged camera, servo, controller, and power modules into a compact enclosure with minimal clearance
- Selected PC-ABS through a material trade study balancing dimensional stability, surface finish, and moldability
- Designed plastic injection components with 8 ejector pins and 4 gating features, and 3D-printed a tooling demonstrator

Traction Battery Pack Cold Plate - Thermal Analysis — **April 2025**

- Designed steel sheet-metal mounts using SOLIDWORKS and FEA to calculating a 3.2 FoS under modeled loads
- Fabricated sheet-metal mounts and machined spacers; produced GD&T drawings to resolve assembly clearances
- Evaluated ANSYS thermal predictions against endurance thermocouple measurements to assess cooling performance
- Reduced peak battery-cell temperature by 6°C during endurance testing by integrating liquid cooling beneath 12 modules

Robotic Rover Antenna Mast - Vibration and Signal Integrity — **March 2026**

- Designed a 9-ft telescoping antenna mast within 1.3 ft stowed height through CAD packaging optimization
- Balanced structural stiffness, mass, and manufacturability through DFM/DFA principles and design trade studies
- Resolved mounting clearance and clamping issues by prototyping 3D-printed PLA brackets and conducting fit checks
- Modeled a 915-MHz rover link over 1 km using MATLAB, estimating 14–19 dB link margin across 1.3–9 ft mast heights

Battery Module Baseplate & Frame – Structural Analysis — **December 2025**

- Designed FR-4 baseplates and a structural frame for a 42-cell EV battery module, integrating cell retention and fuse-links
- Optimized baseplate geometry for manufacturability, applying GD&T to control cell spacing and mounting interfaces
- Analyzed FR-4 battery baseplates under 40g inertial loading using FEA to assess bending stress and deflection

LEADERSHIP & INVOLVEMENT

SJSU Robotics – Mechanical Design & Integration Engineer — **January 2026 – May 2026**
FSAE Formula Spartan Racing – EV Powertrain Mechanical Engineer — **July 2024 – December 2025**
Alpha Phi Omega – Public Relations — **January 2024 – May 2025**